Midterm Project Report

CSS Flexbox, CSS Grid and Responsive Portfolio Design

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Assignment:	Midterm Portfolio	**Submission Date:**	October 25 2026
GitHub Repo:	https://github.com/sergio-aguilar/responsive-portfolio-midterm		
Live Pages URL:	https://sergio-aguilar.github.io/responsive-portfolio-midterm/		

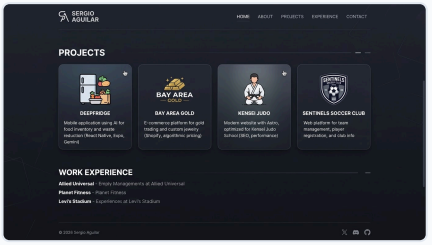
SECTION 1 WHAT CHANGED IN MY PORTFOLIO BEFORE AND AFTER

For this midterm project I took my basic old portfolio layout and turned it into a smooth responsive website. The first version I built in class was super plain and used fixed widths which made it break on smaller screens or phones.

BEFORE Early Version

Basic layout with rigid pixel widths plain floats and no media queries to fix shrinking screens.

Figure 1.1 Original static site before adding Flexbox and CSS Grid



AFTER Updated Portfolio

Modern layout with a Flexbox navbar four card CSS Grid for DeepFridge Bay Area Gold Kensei Judo and Sentinels Soccer Club plus Work Experience.

Figure 1.2 Improved responsive design with fluid grids and media queries

Main things I fixed and added 1 Swapped out old float positioning for a clean Flexbox header bar. 2 Rebuilt the projects section with CSS Grid so cards line up evenly automatically. 3 Added media queries so the site scales smoothly on desktop 1200px tablet 800px and mobile 400px.

SECTION 2 FLEXBOX IMPLEMENTATION

I used Flexbox for the top navigation bar and for aligning elements in one direction. It makes it super easy to push items to opposite sides and keep everything centered vertically without doing weird margin hacks.

1 Navigation Bar Header

The navbar uses display flex to put my logo on the left side and all the page links on the right side with equal spacing.

```
/* Navigation Bar Flexbox Setup */ .navbar { display: flex; justify-content: space-between; /* Pushes logo to left and links to right */ align-items: center; /* Centers logo and links vertically */ flex-wrap: wrap; /* Lets links wrap on small screens */ gap: 16px; /* Spacing between header items */ padding: 16px 24px; background-color: #0f172a; } .nav-links { display: flex; gap: 20px; list-style: none; }
```

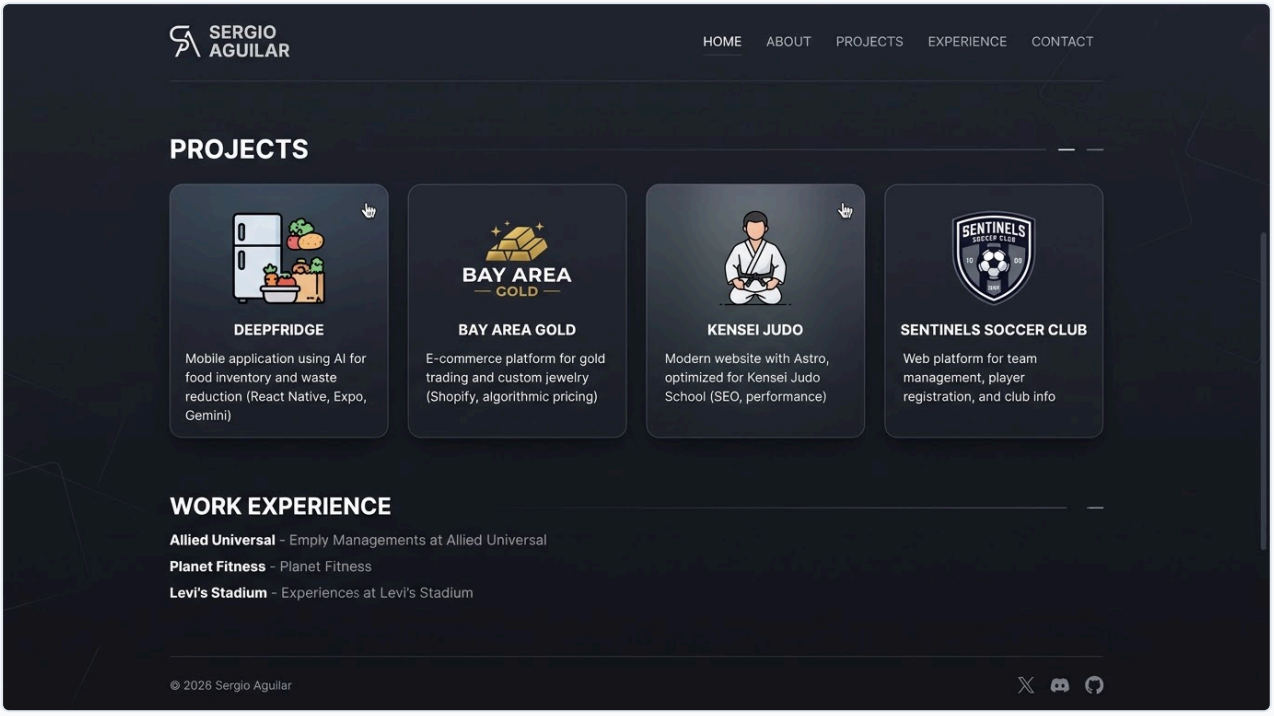


Figure 2.1 Webpage navbar running on Flexbox with space between alignment

How the Flexbox properties work here

- `display flex` turns on flexbox mode for the header bar.
- `justify-content space-between` spreads the title logo and nav links to the far edges.
- `align-items center` keeps everything perfectly lined up straight through the middle.
- `gap 16px` sets a clean space between items so they never crash into each other.

SECTION 3 CSS GRID IMPLEMENTATION

I picked CSS Grid for the projects area because grid is made for two dimensional layouts with rows and columns. It makes building a gallery of cards way easier than using float or manual inline blocks.

Project Gallery Layout

The gallery holds four main project cards DeepFridge Bay Area Gold Kensei Judo and Sentinels Soccer Club.

```
/* CSS Grid Setup for Projects */ .project-gallery { display: grid; grid-template-columns: repeat(auto-fit, minmax(240px, 1fr)); /* Fluid columns */ gap: 20px; /* Space between cards */ padding: 20px 0; } .project-card { background: #1e293b; border-radius: 12px; border: 1px solid #334155; }
```

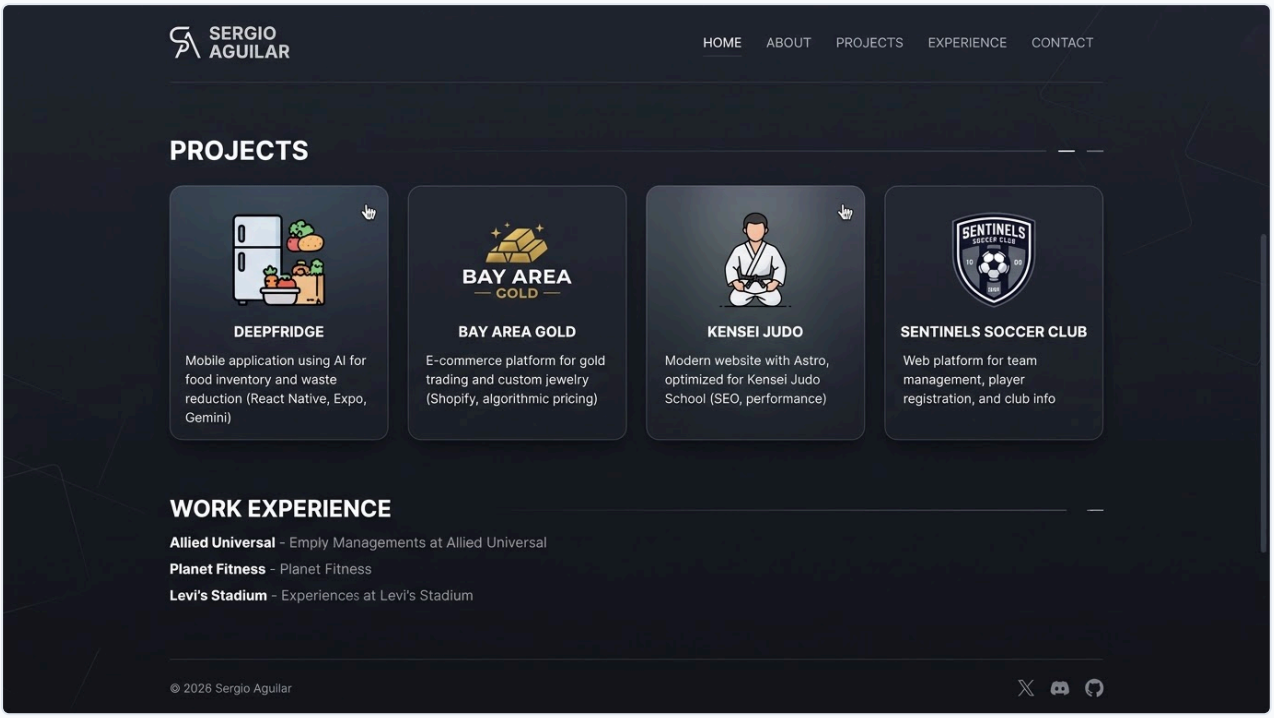


Figure 3.1 CSS Grid displaying four project cards in a neat row

How the Grid properties work here

• `display grid` enables grid controls on the gallery container.

• `repeat auto-fit minmax 240px 1fr` automatically calculates how many columns can fit. It keeps cards at least 240px wide and stretches them evenly with 1fr fraction units.

• `gap 20px` keeps equal spacing between grid rows and columns without adding extra outer margins.

SECTION 4 RESPONSIVE DESIGN AND VIEWPORT BREAKDOWN

I added the viewport meta tag in the head of my HTML file using `<meta name="viewport" content="width=device-width, initial-scale=1.0">` so browsers on phones scale correctly. Then I added two main breakpoints for tablet and mobile screens.

```
/* Media Queries for Responsive Screen Widths */ @media screen and (max-width: 800px) { /* Tablet View */ .navbar { flex-direction: column; align-items: flex-start; } .project-gallery { grid-template-columns: repeat(2, 1fr); } } @media screen and (max-width: 400px) { /* Mobile View */ .project-gallery { grid-template-columns: 1fr; } .nav-links { flex-direction: column; width: 100%; gap: 10px; } }
```

1200px Desktop

Full 4 column grid layout showing DeepFridge Bay Area Gold Kensei Judo and Sentinels Soccer Club across the screen with header links on the right side.

800px Tablet

Grid rearranges into a 2 column setup so cards do not get squished. The navbar stacks links to fit narrower tablet screens.

400px Mobile

Everything stacks into a clean single column flow for phone screens. Work experience items for Allied Universal Planet Fitness and Levis Stadium stack straight down.

How the layout behaves across screens

On big screens 1200px all cards sit side by side in one row. When you resize down to around 800px the tablet rule kicks in and makes it a 2x2 grid. On mobile around 400px it switches to a 1 column stack so nothing overflows sideways and text stays easy to read.

SECTION 5 WRITTEN REFLECTION

Updating my portfolio website for this midterm project was a really good learning experience for working with modern CSS layout tools. I upgraded the overall look by adding a Flexbox header and setting up a CSS Grid gallery for my main projects DeepFridge Bay Area Gold Kensei Judo and Sentinels Soccer Club alongside my Work Experience section featuring Allied Universal Planet Fitness and Levis Stadium.

The trickiest part was getting the grid cards to scale properly on medium screens without text overflowing out of the cards. I solved it by using minmax 240px 1fr combined with media query breakpoints at 800px and 400px to force the grid into fewer columns when the window shrinks.

I chose Flexbox for the top navigation bar and work experience rows because those elements only move in one direction and need easy spacing. I chose CSS Grid for the project gallery because grid handles rows and columns at the same time. While testing on mobile views I learned how important it is to test small screen widths so text does not trigger horizontal scrollbars.

SECTION 6 GITHUB SUBMISSION INFO

My portfolio code images and documentation are pushed to my public GitHub repository.

Requirement	Link or Description
GitHub Repository	`https://github.com/sergio-aguilar/responsive-portfolio-midterm`
Live GitHub Pages	`https://sergio-aguilar.github.io/responsive-portfolio-midterm/`
README file	Contains project details features list live link and instructions for opening the website in a browser.